Failure Analysis Report

FAIL-0693 — Timeout

1. Failure Summary

Failure ID	FAIL-0693
Failure Type	Timeout
Severity	Major
Component	Bluetooth Module
Application	APP-009 — DeviceHub
Build Version	4.5.2
Timestamp	2024-02-29 17:08:26
Environment	ENV-001 — Mobile Lab
OS Version	Android 14
Network	WiFi-6
Resolution	Open

2. Test Case Reference

Test Case ID	TC-1166
Title	Verify that Bluetooth reconnection completes successfully when the battery level...
Execution ID	EXEC-03597
Test Run	RUN-001 — Regression Cycle 1 — Build 4.1.8
Requirement	REQ-0411 — Data Synchronization — Payment Processing

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete playlist shuffle within the defined SLA.
Observed Behaviour	The system exhibited timeout when two concurrent requests are made simultaneously.

4. Error Details

Error Message:

```
Operation timed out after 19561ms; threshold is 8764ms.
```

Stack Trace / Log Excerpt:

```
com.samsung.smarthome.NotificationHandler.execute(line 212)
```

Log Reference	LOG-42752

5. Executed Steps at Time of Failure

1	Disconnect the phone and wait 25 seconds.	The audit log records the event with correct metadata.

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2	Navigate to Settings > Privacy and tap 'Verify Identity'.	The audit log records the event with correct metadata.
3	Simulate high CPU load using the network emulator profile 'WiFi-L...	The UI transitions to the next screen within 3 seconds.
4	Verify that the UI displays 'Sync complete'.	No crash or ANR is observed.

6. Root Cause Analysis

Firmware update binary not validated against checksum before flashing; partial write leaves device in boot loop.

8. Historical Similar Failures

HIST-0080	FAIL-0081	Timeout	5.1.1	2022-04-06	8	Open
HIST-0275	FAIL-0276	Timeout	5.3.8	2023-05-24	9	Wont Fix
HIST-0534	FAIL-0535	Timeout	4.3.6	2022-09-03	13	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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